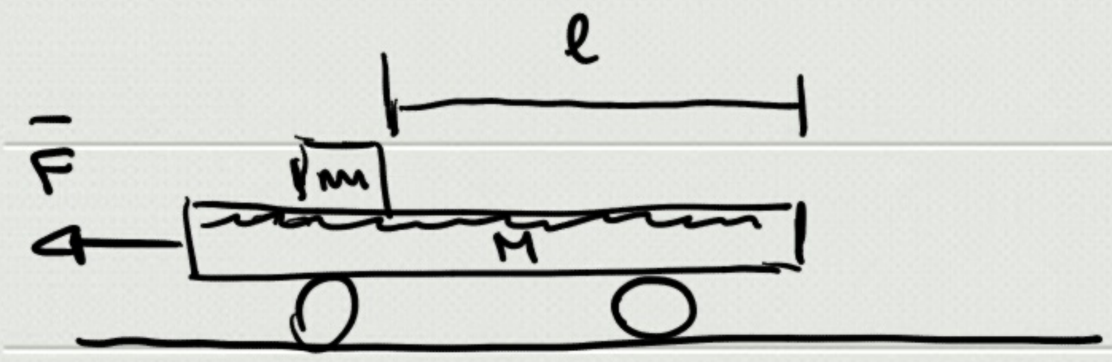
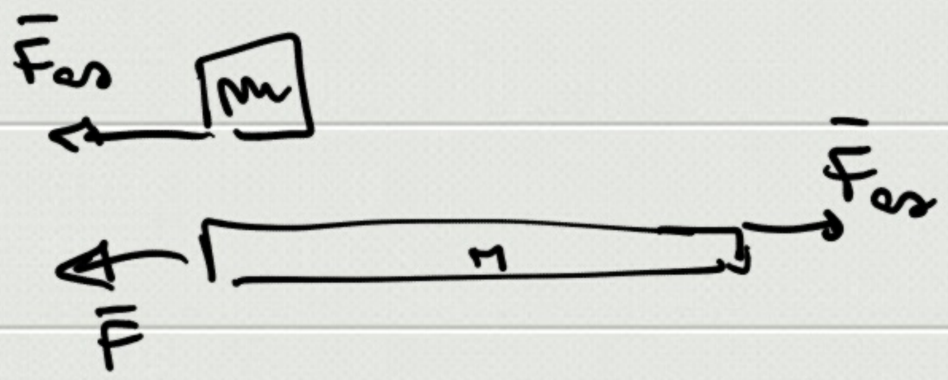


$M = 3\,kg$

$m = 1\,kg$

$l = 3\,m$



$\mu_d = \mu_s = 0.1$

$F = 5\,N$

$$\begin{cases} F_{es} = m a \\ F - F_{es} = M a \end{cases}$$

$$F = (m+M)\,a \Rightarrow a = \frac{F}{m+M}$$

$$\Rightarrow \cancel{m \frac{F}{m+M} \leq \mu_s m g}$$

$1.25\,m/s^2 \quad 0.98\,m/s^2$

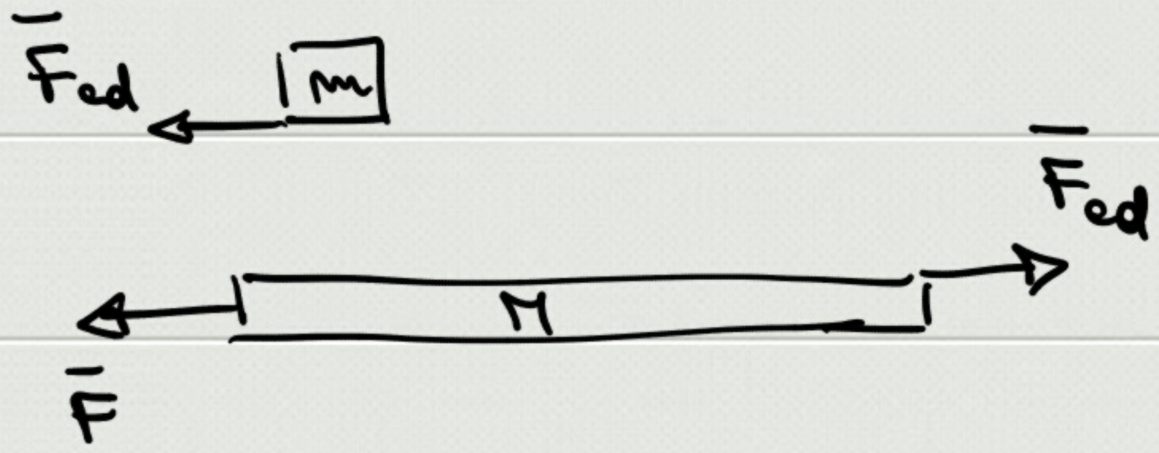
$t_{caduta} = ?$

$$\begin{cases} F - \mu_d m g = M a_M \\ \mu_d m g = m a_m \end{cases} \quad *$$

$$\begin{cases} -F + \mu_d m g = M a_M \\ -\mu_d m g = m a_m \end{cases} \quad *$$

$$l = \frac{1}{2}(a_m - a_M)\,t^2 \quad (a_m - a_M > 0) \quad *$$

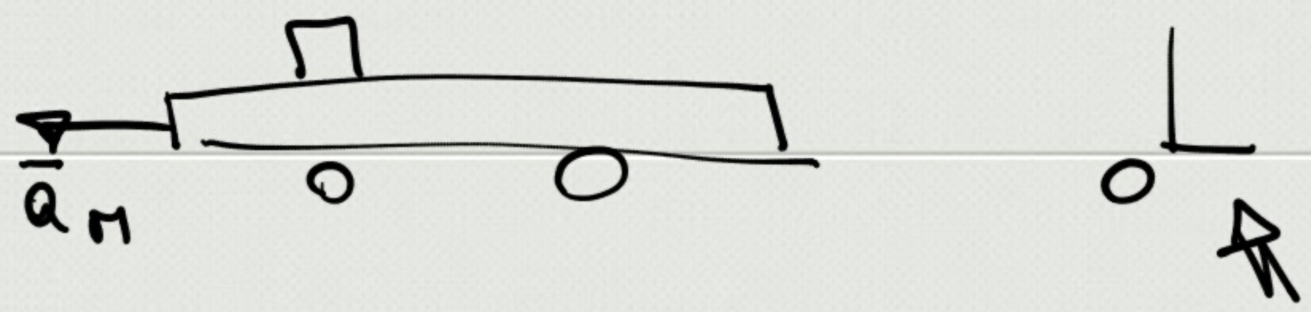
$$l = \frac{1}{2} a_m t^2 \quad (a_m > 0) \quad \underline{\underline{NO}}$$



$$\begin{cases} \mu_d \cancel{m} g = \cancel{m} a_m \\ F - \mu_d m g = M a_M \end{cases}$$

$$\begin{cases} -\mu_d m g = m a_m \\ -F + \mu_d m g = M a_M \end{cases}$$

$$a_m = \mu_d g \qquad a_M = \frac{F - \mu_d m g}{M}$$



$$l = \frac{1}{2} a'_m t^2 \qquad \bar{a} = \bar{a}' + \bar{a}_{o'} + \dots$$

$$\Rightarrow a'_m = a_m - a_M =$$

$$l = \frac{1}{2}(a_m - a_M) t^2 \qquad = -0.35 \ m/s^2$$

$$a'_m = a_m - a_M = 0.35 \ m/s^2$$

$$t = \sqrt{\frac{2l}{|a_m - a_M|}} \cong 4.2 \ s$$